DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)

Project Synopsis

Batch No: F10 **Date: 19 - 07 - 2026**

USN	Name	Email-ID	Mobile Number
4NI23CI064	Nischitha H L	2023ci_nischithahl_b@nie.ac.in	7975750338
4NI23CI068	Omkar S Moodvi	2023ci_omkarsmoodvi_b@nie.ac.in	7676368811
4NI23CI096	Sanmitha H	2023ci_sanmithah_b@nie.ac.in	9353498733

Project Title: Secure and Autonomous Multi-Agent Enterprise Assistant

Introduction: Modern enterprises rely on multiple platforms such as ERP, CRM, HRMS, cloud storage, and project management tools that often operate independently, creating information silos, repetitive tasks, and inefficient collaboration across departments. Traditional AI chatbots provide limited support as they lack workflow coordination, secure access control, and often generate unreliable responses.

This project proposes a Secure Autonomous Multi-Agent Enterprise Workflow Intelligence System that integrates specialized AI agents with LLMs, Retrieval-Augmented Generation (RAG), Zero-Trust Security, Role-Based Access Control (RBAC), Explainable AI, and Human-in-the-Loop validation to deliver secure, accurate, and intelligent enterprise automation.

Objectives:

Design a secure multi-agent architecture for enterprise workflow automation.

Implement intelligent task decomposition using autonomous AI agents.

Integrate Retrieval-Augmented Generation (RAG) for accurate enterprise information retrieval.

Enforce Zero-Trust security with Role-Based Access Control (RBAC).

Reduce AI hallucinations using enterprise knowledge retrieval.

Provide Explainable AI for transparent decision-making.

Enable Human-in-the-Loop validation for sensitive enterprise tasks.

Evaluate system performance using response accuracy, workflow completion time, and security metrics.

Software Requirements – Operating System: Windows 11 / Ubuntu 22.04,
Programming Language: Python 3.11
Frontend: React.js Backend: FastAPI
Database: PostgreSQL / MongoDB, Vector Database: ChromaDB / FAISS
LLM: Llama 3 / Mistral / GPT
LangChain
LangGraph / CrewAI / AutoGen

Hardware Requirements – Intel Core i5/i7 Processor (or equivalent)
16 GB RAM (Minimum 8 GB)
512 GB SSD
Internet Connection
NVIDIA GPU (Optional for local LLM execution)

Abstract: The proposed **Secure Autonomous Multi-Agent Enterprise Workflow Intelligence System** is an AI-driven platform that securely automates enterprise workflows using multiple specialized AI agents. It decomposes complex user requests into subtasks and coordinates agents for information retrieval, analytics, workflow automation, and security verification. The system integrates Retrieval-Augmented Generation (RAG), Zero-Trust Security, and Role-Based Access Control (RBAC) to ensure accurate responses and secure access to enterprise data. Explainable AI and Human-in-the-Loop validation provide transparency and approval for critical operations. Overall, the system improves productivity,

accuracy, security, and scalability for modern enterprise environments.

Literature Survey (Minimum 5 recent research Journals indexed in Scopus/Web of Science):

1. Multi-Agent Systems for Enterprise Workflow Automation.
2. Retrieval-Augmented Generation (RAG) for Knowledge-Intensive AI Applications.
3. Zero-Trust Security Architecture for Enterprise Systems.
4. Explainable Artificial Intelligence (XAI) for Large Language Models.
5. Human-in-the-Loop AI for Enterprise Decision Support.

Existing System & its Drawback:

Existing System

Most organizations use enterprise software such as ERP, CRM, HRMS, Microsoft Copilot, Google Gemini for Workspace, ServiceNow AI, Jira Automation, and traditional AI chatbots to automate routine business operations. These systems help retrieve information, generate reports, answer queries, and manage workflows, but they operate independently and require manual coordination across applications and departments.

Most existing AI assistants rely on a single-agent architecture, where one AI model handles all user requests. Although Retrieval-Augmented Generation (RAG) improves information retrieval, these systems still lack intelligent multi-agent collaboration, dynamic workflow orchestration, secure enterprise-wide automation, and explainable decision-making for complex business processes.

Drawbacks

1. Limited to single-agent architecture.
2. Cannot perform autonomous multi-step workflows.
3. High AI hallucination rate.
4. Weak access control for sensitive enterprise data.
5. Limited explainability and auditability.
6. Poor scalability for complex enterprise environments.

Proposed System & Advantages:

The proposed system introduces multiple autonomous AI agents coordinated through a secure workflow orchestration framework. Specialized agents collaborate to retrieve enterprise knowledge, analyze requests, automate tasks, verify permissions, validate responses, and provide explainable outputs. RAG enhances response accuracy, while Zero-Trust security and RBAC ensure secure enterprise data access.

Advantages

Intelligent multi-agent collaboration.
Dynamic task planning and workflow execution.
Reduced AI hallucinations through RAG.
Strong Zero-Trust security.
Explainable AI with confidence scores.
Human-in-the-Loop validation.
Scalable enterprise architecture.
Faster and more accurate workflow automation.

Guide Details:

Name: Gowtham R Naik

Designation: Assistant Professor

Mobile Number : 78923 42728

College Address: No 50, Koorgalli Village, Hootagalli Industrial Area, Next to BEML, Mysore, Karnataka, 570018